

Pegah Khayatan — PhD Student

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Google Scholar

Education

Sorbonne Université, ISIR (MLIA team)

PhD in Computer Science (expected 2028)

Interpretability, steering and hallucination mitigation in vision-language(-action) models. Advisors: Matthieu Cord, Alasdair Newson.

Paris

2025–2028

Télécom Paris

Engineering Program (GPA: 4.0/4.0)

Palaiseau

2022–2023

École Polytechnique

Engineering Program (Ingénieur Polytechnicien)

Palaiseau

2019–2023

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. Electrical Engineering (ranked 2nd/120) & Mathematics (second major)

Tehran

2017–2019

Publications (* equal contribution)

2026: P. Khayatan, J. Parekh, A. Dapogny, M. Shukor, A. Newson, M. Cord. *When Prompts Override Vision: Prompt-Induced Hallucinations in LVLMS*. **NeurIPS 2026**; also at NeurIPS 2026 Workshops (VLM4RWD). [\[paper\]](#) [\[project\]](#)

2026: P. Khayatan*, S. Meziane*, J. Parekh*, M. Cord. *DiMaS: Distribution Matching for Steering Vision-Language-Action Models*. **ECCV 2026 Workshops** (eXCV). [\[paper\]](#) [\[code\]](#) [\[project\]](#)

2025: J. Parekh*, P. Khayatan*, M. Shukor, A. Dapogny, A. Newson, M. Cord. *Learning to Steer: Input-dependent Steering for Multimodal LLMs*. **NeurIPS 2025**. [\[paper\]](#) [\[code\]](#)

2025: P. Khayatan*, M. Shukor*, J. Parekh*, A. Dapogny, M. Cord. *Analyzing Finetuning Representation Shift for Multimodal LLMs Steering*. **ICCV 2025**. [\[paper\]](#) [\[code\]](#)

2024: J. Parekh, P. Khayatan, M. Shukor, A. Newson, M. Cord. *A Concept-Based Explainability Framework for Large Multimodal Models*. **NeurIPS 2024**. [\[paper\]](#) [\[code\]](#)

2024: Y. Yin, P. Khayatan, É. Zablocki, A. Boulch, M. Cord. *ReGentS: Real-World Safety-Critical Driving Scenario Generation Made Stable*. **ECCV 2024 Workshops** (W-CODA). [\[paper\]](#) [\[code\]](#)

Research Experience

Sorbonne Université, ISIR (MLIA team)

Research Engineer (supervisor: Matthieu Cord)

Paris

Mar 2024 – Sep 2025

- Concept-based explainability of large multimodal models via dictionary learning (NeurIPS 2024).
- Analysis of fine-tuning representation shift and training-free concept steering of MLLMs (ICCV 2025).
- Input-dependent steering with a lightweight learned steerer for safety and hallucination mitigation (NeurIPS 2025).

Valeo.ai

Research Intern (supervisors: Éloi Zablocki, Alexandre Boulch)

Paris

Jul 2023 – Jan 2024

Adversarial generation of safety-critical driving scenarios for motion-planner stress testing, with a differentiable simulator (ECCV-W 2024). [\[code\]](#)

EPFL, VITA Lab

Research Intern (supervisors: Alexandre Alahi, Taylor Mordan)

Lausanne

Mar 2022 – Aug 2022

Clothing keypoint localization and category classification; released an OpenPifPaf plugin for DeepFashion. [\[code\]](#)

Amadeus

Research Intern – ML-based intrusion detection for cloud security

Jun 2021 – Aug 2021

Skills & Languages

Programming: Python, PyTorch, Hugging Face Transformers, C++, Git, Linux, SLURM / HPC clusters

Research: Vision-language models, interpretability (dictionary learning, concepts), activation steering, preference optimization (DPO)

Languages: Persian (native), English (fluent, TOEFL 105), French (fluent, TCF C1)

Honors, Teaching & Outreach

2023: Represented Télécom Paris at SWERC (ICPC regional programming contest), Milan.

2019–2022: Eiffel Excellence Scholarship (French Ministry for Europe and Foreign Affairs); Fulgence Bienvenüe Scholarship (École Polytechnique).

2019–2021: École Polytechnique: TA in Probability Theory (2020); Fun MOOC math content for high-schoolers.

2016–2017: Bronze Medal, IMC Global Competition (Iran's team, 2016); top 0.002% in Iranian University Entrance Exam (2017).